

Generation Interconnection System Impact Study Report

For Queue Project Q23-2322

New Salisbury 138 kV

May 16, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2322. The Interconnection Customer proposes a 73.0 MW solar photovoltaic facility interconnecting at the New Salisbury 138 kV substation in the Virginia area. The request was received on June 19, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 670 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$174,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: DPC PRAIRIE DU CHIEN – CHERRY 138.0 kV

Contingency: CHATOM – PURDY 138.0 kV

Loading: 108.1% (Rating: 264 MVA)

DFAX: 11.6%

Pre-project loading: 104.9%

2. Facility: CHERRY – MOUNT JACKSON 138.0 kV

Contingency: PURDY – CHATOM 138.0 kV

Loading: 107.8% (Rating: 254 MVA)

DFAX: 11.6%

Pre-project loading: 104.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DPC PRAIRIE DU CHIEN – CHERRY 138.0 kV	Reconductor line	\$1,932,000	5% (\$98,000)
CHERRY – MOUNT JACKSON 138.0 kV	Reconductor line	\$1,493,000	5% (\$76,000)
	Total allocated to project		\$174,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.